

PROJECT DOCUMENTATION

FARM MANAGEMENT SYSTEM

A Salesforce-Based Application for Managing Farmers, Farms, Crops and Buyers

Platform: Salesforce Lightning Experience (Developer Edition)

Custom Objects: Farmer__c, Farm__c, Crop__c, Buyer__c

Application: Farm Management System (Lightning App)

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1. Abstract

This document presents the design, configuration and verification of the Farm Management System, a Salesforce-based application built on the Lightning Experience platform in a Salesforce Developer Edition org. The system replaces scattered, manually maintained records with a single, structured data store for four business entities: farmers, farms, crops and buyers.

The solution is built entirely from declarative Salesforce configuration. Four custom objects (Farmer__c, Farm__c, Crop__c and Buyer__c) hold the data; a master-detail relationship links each farm to its farmer, and a lookup relationship links each crop to its farm. A dependent picklist ties the district of a farmer to the selected state, and a currency formula field calculates the expected yield per acre for every crop. Users work in a dedicated Lightning App named Farm Management System, which exposes the four object tabs together with Reports and Dashboards.

Access to the application is controlled through the System Administrator, Farm Manager and Sales Department profiles, a role hierarchy containing the Farm Manager and Sales Department roles, two corresponding users, and two criteria-based sharing rules. Three reports (Farmers with Farms, Crops Report and Buyers) and one dashboard (Farmers With farms) summarise the stored data. Every statement in this document is supported by screenshots captured from the working org, which appear as numbered figures.

2. Introduction

2.1 Project Overview

Agricultural operations generate a continuous stream of information: who owns and operates each farm, what is cultivated, how much is harvested, and to whom produce can be sold. When this information is kept in registers or separate spreadsheets, it is difficult to keep consistent, search or summarise. The Farm Management System addresses this by modelling the information on the Salesforce platform, where it is stored in the cloud, related across objects, and reported on in real time.

The application manages four kinds of records: Farmers, Farms, Crops and Buyers. It is delivered as a Lightning App so that all four record types, together with reports and dashboards, are available from one navigation bar.

2.2 Purpose

The purpose of the project is to provide a structured, searchable and shareable record-keeping system for farm data. Specifically, it is intended to:

- store farmer, farm, crop and buyer information in one centralised Salesforce org;
- relate crops to the farms on which they are grown, and farms to the farmers who operate them;
- present the stored data through list views, reports and a dashboard; and
- restrict access to the application and to selected records by profile, role and sharing rule.

3. Problem Statement

Farm-related records are commonly maintained on paper or in disconnected files. This leads to the following problems for the people who rely on them:

Table 1: Problems addressed by the Farm Management System

User	Problem	Consequence
Farmer	Farm and crop details are written in registers and are easily incomplete or incorrect.	Records are unreliable and hard to update.
Farm manager	Information on farms, crops and yields is scattered across files and sheets.	Summaries are slow to prepare and may be inaccurate.
Buyer	There is no single, current list of buyers and their contact details.	Buyers are difficult to identify, classify and contact.
System administrator	Access to records is not systematically controlled.	Data may be seen or changed by users who should not have access.

The Farm Management System responds to these problems by keeping every record in one place, relating records to each other, and applying Salesforce security features to control who can use the application.

4. Objectives

The objectives of the project are:

- To create the custom objects Farmer, Farm, Crop and Buyer with the fields needed to describe each entity.
- To define relationships so that each farm belongs to a farmer and each crop is recorded against a farm.
- To restrict the district choices of a farmer according to the selected state by using a field dependency.
- To calculate the expected yield per acre of each crop automatically with a formula field.
- To provide a single Lightning App with tabs for the four objects, Reports and Dashboards.
- To configure profiles, roles, users and sharing rules for the Farm Manager and Sales Department functions.
- To provide reports and a dashboard that summarise farms, crops and buyers.

5. Requirements

5.1 Stakeholders

The application is intended for farmers whose data is recorded, farm managers who maintain farm and crop data and review reports, sales department users who work with buyer data, and system administrators who configure and secure the org. The Farm Manager and Sales Department functions are represented in the org by the custom profiles, roles and users described in Section 16.

5.2 Functional Requirements

Table 2: Functional requirements

ID	Requirement	Implemented through
FR-1	Maintain farmer records: name, contact number, state, district and village.	Farmer__c object
FR-2	Maintain farm records: name, farm type, location, size, soil quality, irrigation type and owning farmer.	Farm__c object; master-detail field Farmer
FR-3	Maintain crop records: name, farm, planting and harvest dates, acres harvested, actual yield and expected yield per acre.	Crop__c object; lookup field Farm; formula field
FR-4	Maintain buyer records: name, type, contact, email and address.	Buyer__c object
FR-5	Offer only the districts that correspond to the selected state.	Field dependency (State controls District)
FR-6	Provide a single application for navigating all modules.	Lightning App: Farm Management System
FR-7	Control which profiles can open the application.	App User Profiles; custom profiles
FR-8	Share Crop records with the Farm Manager role and Buyer records with the Sales Department role.	Criteria-based sharing rules
FR-9	Report on farms by irrigation type, crops by name and buyers by type.	Three reports
FR-10	Visualise report data in a dashboard.	Dashboard: Farmers With farms

5.3 Non-Functional Requirements

Table 3: Non-functional requirements

Attribute	Description
Usability	Users work through the standard Salesforce Lightning interface, with object tabs, list views with search, and a navigation bar.
Security	Access is governed by profiles, roles, role-based users and sharing rules.
Data quality	Picklist fields offer predefined values, the record name of each object is a required field, and the formula field removes the need to calculate expected yield by hand.
Availability and scalability	The application is hosted on the Salesforce cloud platform, so the system can grow in records and users within the limits of the org edition.

5.4 Technology Stack

Table 4: Technology stack

Layer	Technology
User interface	Salesforce Lightning Experience
Application platform	Salesforce Platform (Developer Edition org)
Data storage	Salesforce custom objects (Farmer__c, Farm__c, Crop__c, Buyer__c)
Data logic	Master-detail and lookup relationships, dependent picklist, formula field
Security	Profiles, roles, users and sharing rules
Analytics	Salesforce Reports and Dashboards

6. Proposed System

The proposed system is a Salesforce Lightning App, named Farm Management System, built on four custom objects. Instead of recording information in separate files, users create records for farmers, farms, crops and buyers in one org. Each record type has its own tab, list view and record form, and the records are connected as follows:

- **Farmer → Farm:** each farm record must be linked to a farmer through a master-detail relationship.
- **Farm → Crop:** each crop record is linked to a farm through a lookup relationship.
- **Buyer:** buyer records are maintained as a standalone list with type and contact details.

Reports built on these records are shown on a dashboard in the same application, so that the person managing the farms can move from data entry to analysis without leaving the app.

Table 5: Summary of the proposed system

Aspect	Description
Problem	Farm, crop, farmer and buyer information is difficult to keep complete, consistent and summarised when held in separate manual records.
Solution	A centralised Salesforce application with four related custom objects, a Lightning App, reports and a dashboard.
Controls	Dependent picklist, formula field, profiles, roles and sharing rules.
Outcome	Searchable list views for each module, report totals by irrigation type, crop and buyer type, and a dashboard that visualises the same data.

7. System Architecture

The system follows the layered structure of a Salesforce application. Each layer below corresponds to a component that is shown in the supplied screenshots.

Table 6: Architecture layers

Layer	Components	Reference
Presentation	Lightning App “Farm Management System” with the Farms, Farmers, Crops, Buyers, Reports and Dashboards navigation items; list views; record forms	Figures 5–9, 14–17

Layer	Components	Reference
Application logic	Master-detail and lookup relationships; State–District field dependency; Expected Yield per Acre formula	Figures 6, 10–13
Data	Custom objects Farmer__c, Farm__c, Crop__c and Buyer__c	Figures 1–4
Security	Profiles, role hierarchy, users, sharing rules	Figures 18–23
Analytics	Reports (Farmer with Farms, Crops Report, Buyers) and the dashboard Farmers With farms	Figures 24–27

7.1 Data Flow

A user signs in to Salesforce, opens the Farm Management System app and creates or edits records in the object tabs. The records are stored in the custom objects and are connected by the relationship fields. Reports read the stored records and group them (by irrigation type, crop name or buyer type), and the dashboard displays report results as a gauge, a bar chart and a donut chart.

Table 7: Data flow

Step	Activity
1	User opens the Farm Management System app.
2	User creates or updates Farmer, Farm, Crop or Buyer records.
3	Records are stored in the corresponding custom objects and linked through relationship fields.
4	Reports retrieve and group the stored records.
5	The dashboard presents report data as charts.

8. Salesforce Architecture

The application is configured in a Salesforce Developer Edition org; the “Developer Edition” label is visible in the header of every supplied screenshot. All configuration is carried out through Setup, mainly in Object Manager, Tabs, the Lightning App Builder, Profiles, Roles, Users and Sharing Settings.

Table 8: Salesforce components used in the project

Component	Use in the project
Custom objects	Farmer__c, Farm__c, Crop__c and Buyer__c store the application data.
Custom fields	Text, picklist, phone, email, text area, date, number, currency and formula fields describe each entity.
Relationships	Master-detail (Farm to Farmer) and lookup (Crop to Farm).
Field dependency	State (controlling field) and District (dependent field) on Farmer.
Custom tabs	Buyers, Crops, Farmers and Farms tabs give access to the object records.
Lightning App	Farm Management System groups the object tabs, Reports and Dashboards.
Profiles	Farm Manager and Sales Department (custom) and System Administrator are assigned to the app.
Roles and users	Farm Manager and Sales Department roles, each with a corresponding user.

Component	Use in the project
Sharing rules	Criteria-based rules for Crop (shared with the Farm Manager role) and Buyer (shared with the Sales Department role).
Reports and dashboard	Three reports and one dashboard with three components.

9. Data Model / Object Structure

The data model consists of four custom objects. Figures 1 to 4 show the Details page of each object in Object Manager. All four objects are custom, are deployed, and have Enable Reports selected. Track Field History is selected for the Farm object only, and Track Activities is not selected for any of the objects.

Table 9: Custom objects

Object	API name	Singular label	Plural label	Status
Farmer	Farmer__c	Farmer	Farmers	Deployed
Farm	Farm__c	Farm	Farms	Deployed
Crop	Crop__c	Crop	Crops	Deployed
Buyer	Buyer__c	Buyer	Buyers	Deployed

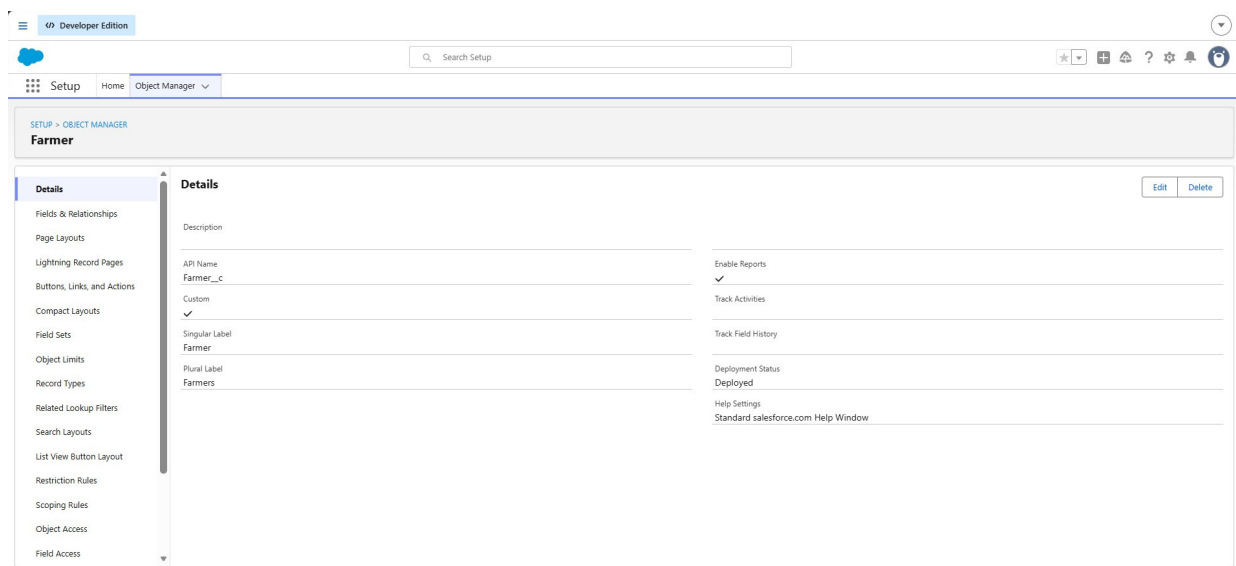


Figure 1: Farmer object details in Object Manager (API name Farmer__c)

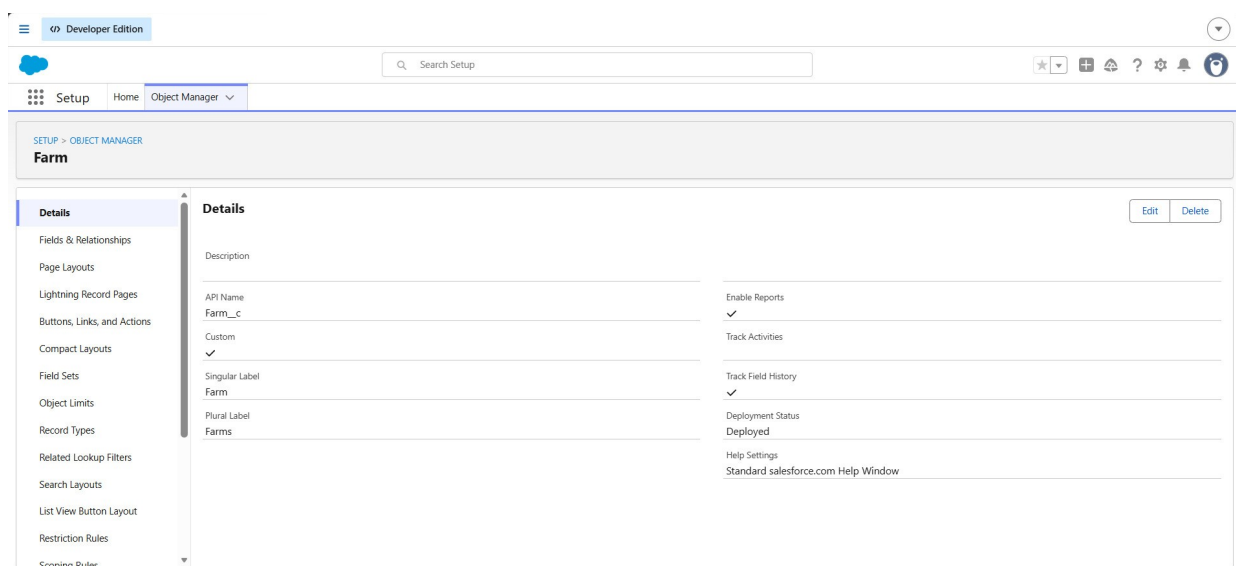


Figure 2: Farm object details in Object Manager (API name Farm__c, Track Field History enabled)

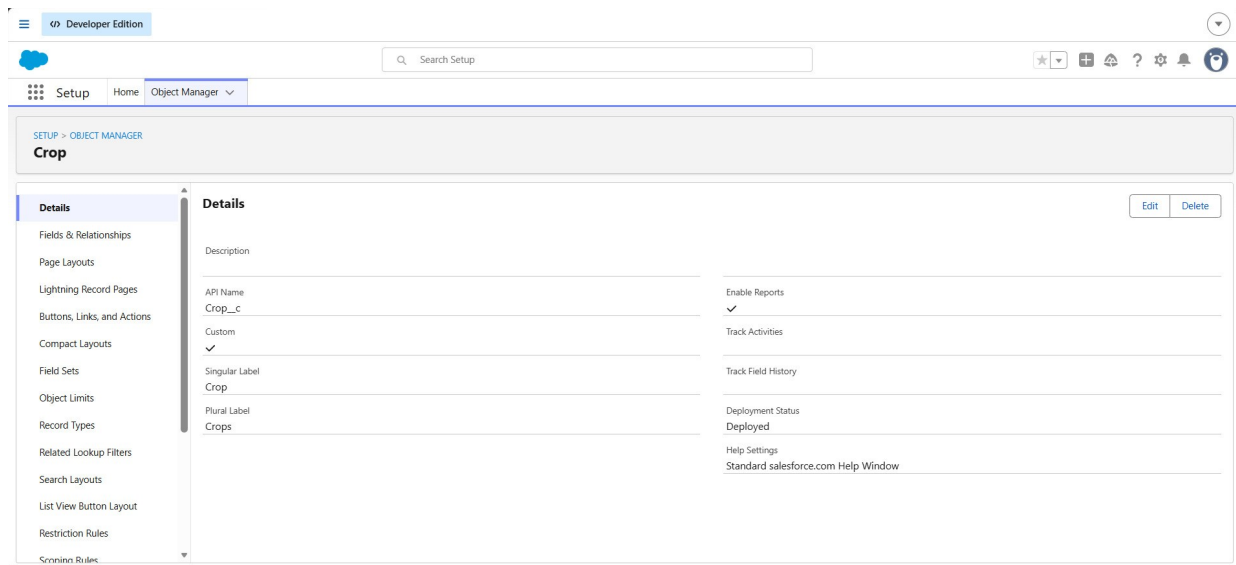


Figure 3: Crop object details in Object Manager (API name Crop__c)

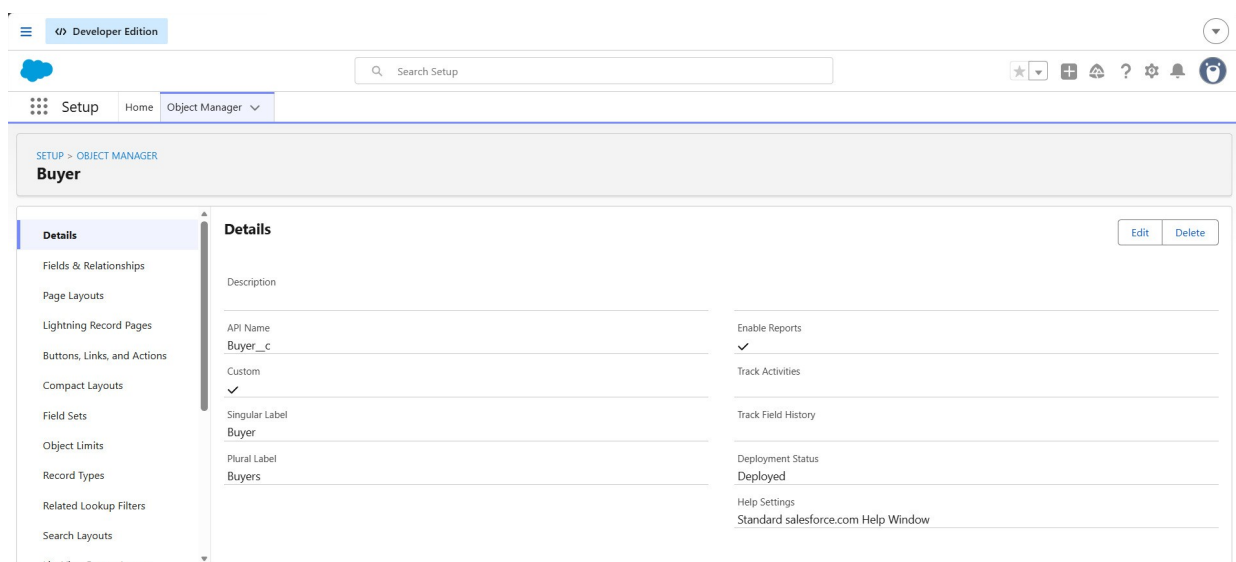


Figure 4: Buyer object details in Object Manager (API name Buyer__c)

9.1 Object Relationships

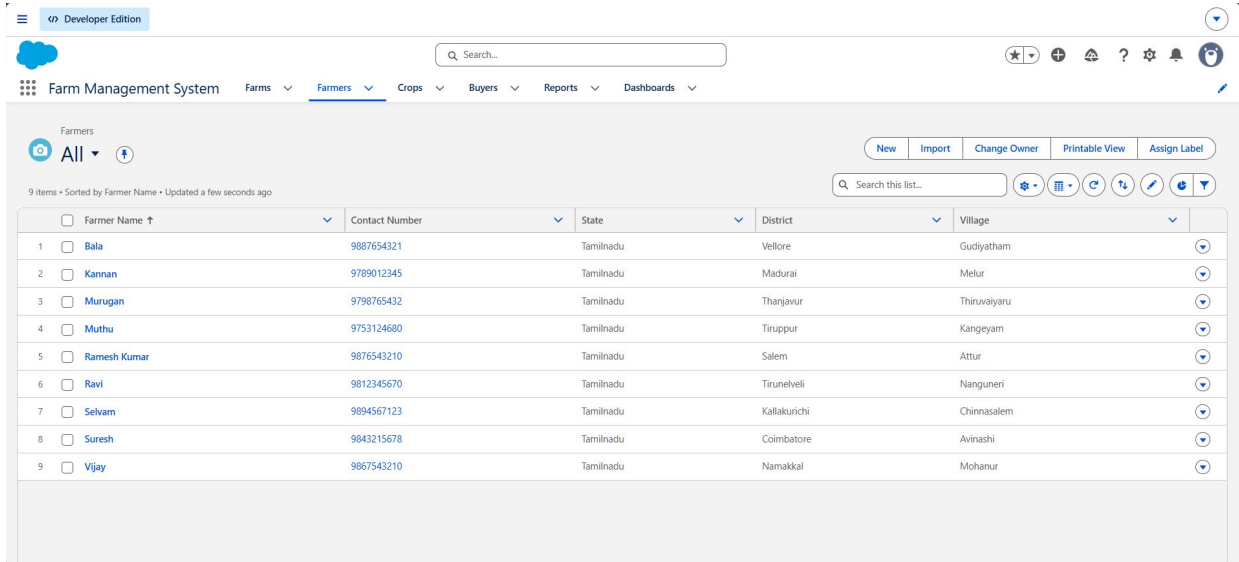
Table 10: Relationships between objects

Child object	Field	Parent object	Type
Farm__c	Farmer (Farmer__c)	Farmer__c	Master-Detail
Crop__c	Farm (Farm__c)	Farm__c	Lookup
Buyer__c	—	None	No relationship to the other three objects is shown

The resulting structure is Farmer (one) to Farm (many) to Crop (many). In a master-detail relationship the detail record cannot exist without its master, which is why the Farm object has no separate Owner field and inherits its sharing from the Farmer record.

10. Farmer Management

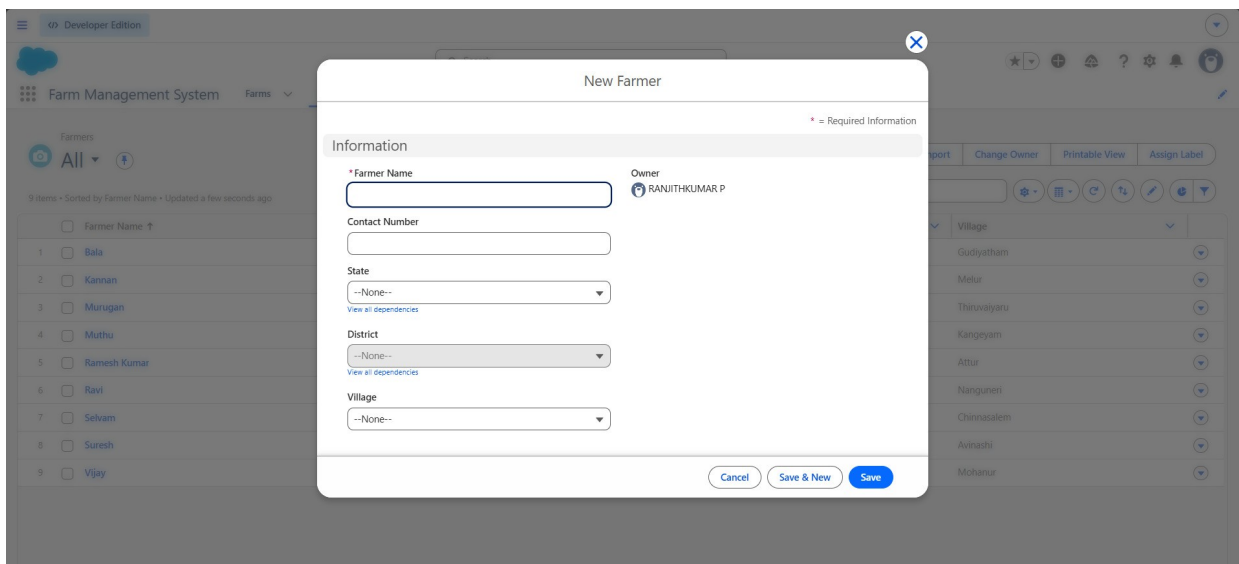
The Farmer module stores the people who operate the farms. The Farmers tab of the Farm Management System app displays the Farmer records in the “All” list view (Figure 5). At the time of capture the list contained nine farmers, sorted by Farmer Name, with the columns Farmer Name, Contact Number, State, District and Village. All nine records show Tamilnadu as the state.



	Farmer Name	Contact Number	State	District	Village
1	Bala	9887654321	Tamilnadu	Vellore	Gudiyatham
2	Kannan	9789012345	Tamilnadu	Madurai	Melur
3	Murugan	9798765432	Tamilnadu	Thanjavur	Thiruvaiyaru
4	Muthu	9753124680	Tamilnadu	Tiruppur	Kangeyam
5	Ramesh Kumar	9876543210	Tamilnadu	Salem	Attur
6	Ravi	9812345670	Tamilnadu	Tirunelveli	Nanguneri
7	Selvam	9894567123	Tamilnadu	Kallakurichi	Chinnasalem
8	Suresh	9843215678	Tamilnadu	Coimbatore	Avinashi
9	Vijay	9867543210	Tamilnadu	Namakkal	Mohanur

Figure 5: Farmers tab in the Farm Management System app showing the All list view

New farmers are created from the New button of the list view. The New Farmer form (Figure 6) shows Farmer Name as a required field, followed by Contact Number, State, District and Village. The Owner of the record is set to the current user. Below State and District, the form displays the link “View all dependencies”, and the District picklist is disabled until a State is chosen, which demonstrates the field dependency described in Section 14.



New Farmer

* = Required Information

Information

* Farmer Name

Contact Number

State

[View all dependencies](#)

District

[View all dependencies](#)

Village

Owner: RANJITHKUMAR P

[Cancel](#) [Save & New](#) [Save](#)

Figure 6: New Farmer record form showing the required Farmer Name and the State, District and Village picklists

Table 11: Farmer records shown in the list view

Farmer Name	State	District	Village
Bala	Tamilnadu	Vellore	Gudiyatham

11. Farm Management

The Farm module records the farms operated by farmers. Each farm has a Farm Type, a Location, a Size, a Soil Quality and an Irrigation Type, and is linked to a farmer. Figure 7 shows the Farms tab with six farm records sorted by Farms Name.

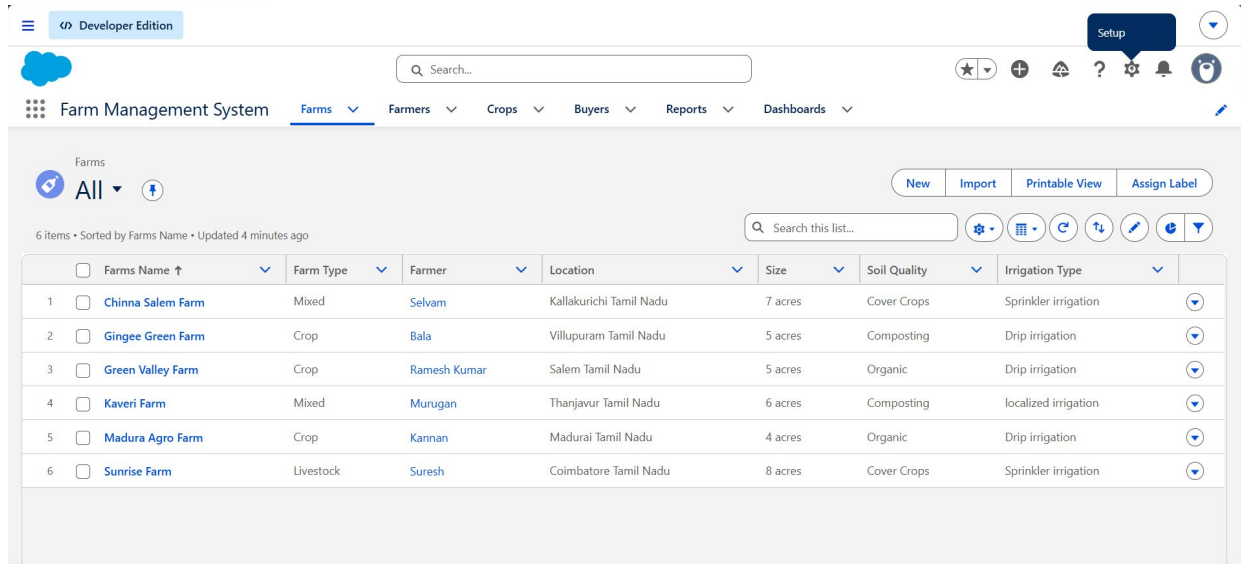


Figure 7: Farms tab showing the All list view with farm type, farmer, location, size, soil quality and irrigation type

Table 12: Farm records shown in the list view

Farm	Type	Farmer	Location	Size	Soil quality	Irrigation
Chinna Salem Farm	Mixed	Selvam	Kallakurichi Tamil Nadu	7 acres	Cover Crops	Sprinkler irrigation
Gingee Green Farm	Crop	Bala	Villupuram Tamil Nadu	5 acres	Composting	Drip irrigation
Green Valley Farm	Crop	Ramesh Kumar	Salem Tamil Nadu	5 acres	Organic	Drip irrigation
Kaveri Farm	Mixed	Murugan	Thanjavur Tamil Nadu	6 acres	Composting	localized irrigation
Madura Agro Farm	Crop	Kannan	Madurai Tamil Nadu	4 acres	Organic	Drip irrigation
Sunrise Farm	Livestock	Suresh	Coimbatore Tamil Nadu	8 acres	Cover Crops	Sprinkler irrigation

The Farmer column is the master-detail field that connects each farm to a Farmer record. The Farm object is also the only one of the four objects for which Track Field History is enabled (Figure 2).

12. Crop Management

The Crop module records what is grown on each farm and the associated planting, harvest and yield information. Figure 8 shows the Crops tab with five records sorted by Crop Name. Each crop is linked to a farm through the Farm lookup field.

Crop Name	Farm	Total Acres Harvested	Actual Yield	Expected Yield per Acre	Planting Date	Harvest date
Groundnut	Madura Agro Farm	4	₹18,000	₹4,500.00	20/06/2026	25/09/2026
Maize	Sunrise Farm	8	₹32,000	₹4,000.00	10/07/2026	20/11/2026
Rice	Gingee Green Farm	5	₹24,000	₹4,800.00	10/06/2026	10/10/2026
Rice	Green Valley Farm	5	₹25,000	₹5,000.00	15/06/2026	15/10/2026
Sugarcane	Kaveri Farm	6	₹45,000	₹7,500.00	05/01/2026	05/12/2026

Figure 8: Crops tab showing the All list view with farm, acres harvested, actual yield, expected yield per acre and dates

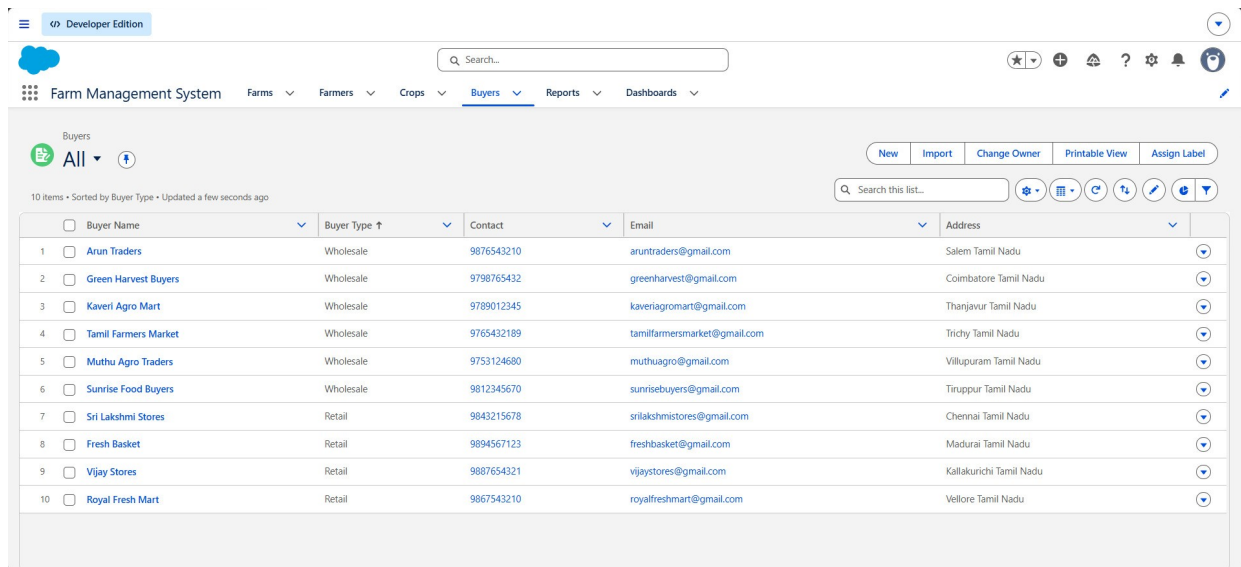
Table 13: Crop records shown in the list view

Crop	Farm	Acres harvested	Actual yield	Expected yield per acre	Planting date	Harvest date
Groundnut	Madura Agro Farm	4	₹18,000	₹4,500.00	20/06/2026	25/09/2026
Maize	Sunrise Farm	8	₹32,000	₹4,000.00	10/07/2026	20/11/2026
Rice	Gingee Green Farm	5	₹24,000	₹4,800.00	10/06/2026	10/10/2026
Rice	Green Valley Farm	5	₹25,000	₹5,000.00	15/06/2026	15/10/2026
Sugarcane	Kaveri Farm	6	₹45,000	₹7,500.00	05/01/2026	05/12/2026

The Expected Yield per Acre column is a formula field. For every record in Figure 8 the displayed value equals Actual Yield divided by Total Acres Harvested (for example, $18,000 \div 4 = 4,500$). Actual Yield is defined as a Currency field, so its values are displayed in rupees.

13. Buyer Management

The Buyer module maintains the list of buyers, classified as Wholesale or Retail, with their contact number, email address and postal address. Figure 9 shows the Buyers tab with ten records sorted by Buyer Type.



☐	Buyer Name	Buyer Type	Contact	Email	Address
1	Arun Traders	Wholesale	9876543210	aruntraders@gmail.com	Salem Tamil Nadu
2	Green Harvest Buyers	Wholesale	9798765432	greenharvest@gmail.com	Coimbatore Tamil Nadu
3	Kaveri Agro Mart	Wholesale	9789012345	kaveriagromart@gmail.com	Thanjavur Tamil Nadu
4	Tamil Farmers Market	Wholesale	9765432189	tamilfarmersmarket@gmail.com	Trichy Tamil Nadu
5	Muthu Agro Traders	Wholesale	9753124680	muthuagro@gmail.com	Villupuram Tamil Nadu
6	Sunrise Food Buyers	Wholesale	9812345670	sunrisebuyers@gmail.com	Tiruppur Tamil Nadu
7	Sri Lakshmi Stores	Retail	9843215678	srilakshmistores@gmail.com	Chennai Tamil Nadu
8	Fresh Basket	Retail	9894567123	freshbasket@gmail.com	Madurai Tamil Nadu
9	Vijay Stores	Retail	9887654321	vijaystores@gmail.com	Kallakurichi Tamil Nadu
10	Royal Fresh Mart	Retail	9867543210	royalfreshmart@gmail.com	Vellore Tamil Nadu

Figure 9: Buyers tab showing the All list view with buyer type, contact, email and address

Table 14: Buyer records shown in the list view

Buyer	Type	Address
Arun Traders	Wholesale	Salem Tamil Nadu
Green Harvest Buyers	Wholesale	Coimbatore Tamil Nadu
Kaveri Agro Mart	Wholesale	Thanjavur Tamil Nadu
Tamil Farmers Market	Wholesale	Trichy Tamil Nadu
Muthu Agro Traders	Wholesale	Villupuram Tamil Nadu
Sunrise Food Buyers	Wholesale	Tiruppur Tamil Nadu
Sri Lakshmi Stores	Retail	Chennai Tamil Nadu
Fresh Basket	Retail	Madurai Tamil Nadu
Vijay Stores	Retail	Kallakurichi Tamil Nadu
Royal Fresh Mart	Retail	Vellore Tamil Nadu

14. Fields and Relationships

Fields are the columns of a Salesforce object. Figures 10 to 13 show the Fields & Relationships page of each object in Object Manager. The tables that accompany each figure list the fields exactly as they appear in the screenshots; Created By and Last Modified By are standard system fields.

14.1 Farmer Fields

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Contact Number	Contact_Number__c	Phone		
Created By	CreatedById	Lookup(User)		
District	District__c	Picklist	State	
Farmer Name	Name	Text(80)		✓
Last Modified By	LastModifiedById	Lookup(User)		
Owner	OwnerId	Lookup(User,Group)		✓
State	State__c	Picklist		
Village	Village__c	Picklist		

Figure 10: Farmer object: Fields & Relationships (8 items)

Table 15: Fields of Farmer__c

Field label	Field name	Data type	Notes
Farmer Name	Name	Text(80)	Record name; indexed
Contact Number	Contact_Number__c	Phone	
State	State__c	Picklist	Controlling field for District
District	District__c	Picklist	Controlling field: State
Village	Village__c	Picklist	
Owner	OwnerId	Lookup(User,Group)	Indexed
Created By	CreatedById	Lookup(User)	System field
Last Modified By	LastModifiedById	Lookup(User)	System field

14.2 Farm Fields

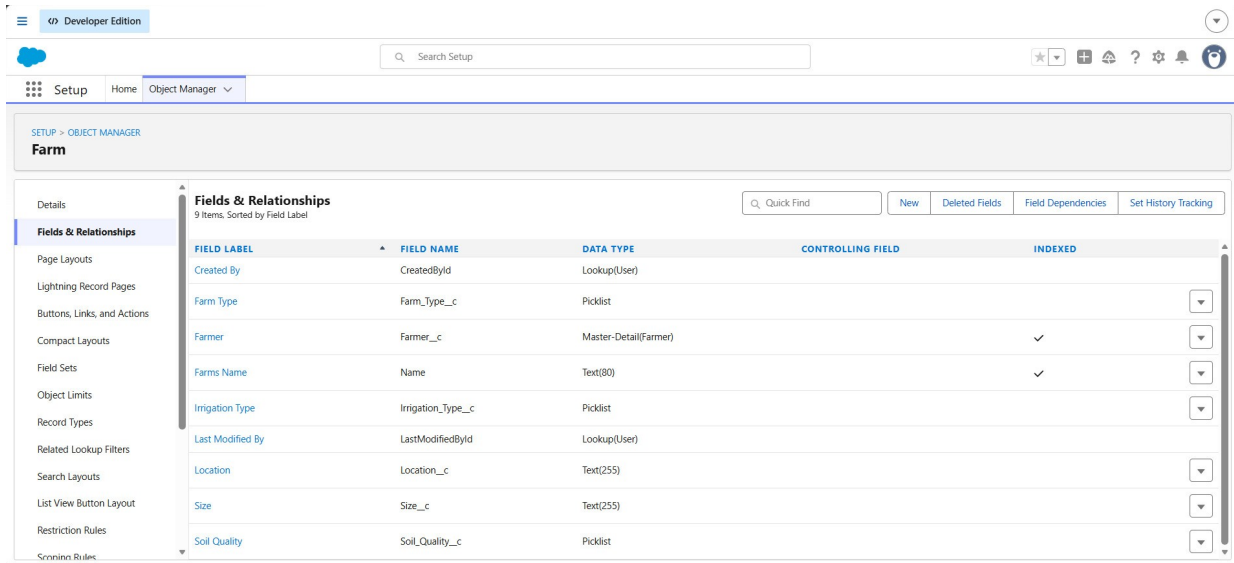


Figure 11: Farm object: Fields & Relationships (9 items)

Table 16: Fields of Farm__c

Field label	Field name	Data type	Notes
Farms Name	Name	Text(80)	Record name; indexed
Farmer	Farmer__c	Master-Detail(Farmer)	Relationship to Farmer; indexed
Farm Type	Farm_Type__c	Picklist	Values seen in records: Crop, Livestock, Mixed
Location	Location__c	Text(255)	
Size	Size__c	Text(255)	Entered as text, for example “7 acres”
Soil Quality	Soil_Quality__c	Picklist	Values seen in records: Organic, Cover Crops, Composting
Irrigation Type	Irrigation_Type__c	Picklist	Values seen in records: Sprinkler irrigation, Drip irrigation, localized irrigation
Created By	CreatedById	Lookup(User)	System field
Last Modified By	LastModifiedById	Lookup(User)	System field

14.3 Crop Fields

Figure 12 shows the 'Crop' object in the 'Object Manager' section of the 'Developer Edition' interface. The 'Fields & Relationships' tab is selected, displaying a table of 10 fields. The table has columns for Field Label, Field Name, Data Type, Controlling Field, and Indexed. The fields are sorted by Field Label.

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Actual Yield	Actual_Yield__c	Currency(18, 0)		
Created By	CreatedById	Lookup(User)		
Crop Name	Name	Text(80)		✓
Expected Yield per Acre	Expected_Yield_per_Acre__c	Formula (Currency)		
Farm	Farm__c	Lookup(Farm)		✓
Harvest date	Harvest_date__c	Date		
Last Modified By	LastModifiedById	Lookup(User)		
Owner	OwnerId	Lookup(User,Group)		✓
Planting Date	Planting_Date__c	Date		
Total Acres Harvested	Total_Acres_Harvested__c	Number(18, 0)		

Figure 12: Crop object: Fields & Relationships (10 items)

Table 17: Fields of Crop__c

Field label	Field name	Data type	Notes
Crop Name	Name	Text(80)	Record name; indexed
Farm	Farm__c	Lookup(Farm)	Relationship to Farm; indexed
Planting Date	Planting_Date__c	Date	
Harvest date	Harvest_date__c	Date	
Total Acres Harvested	Total_Acres_Harvested__c	Number(18, 0)	
Actual Yield	Actual_Yield__c	Currency(18, 0)	
Expected Yield per Acre	Expected_Yield_per_Acre__c	Formula (Currency)	Calculated field
Owner	OwnerId	Lookup(User,Group)	Indexed
Created By	CreatedById	Lookup(User)	System field
Last Modified By	LastModifiedById	Lookup(User)	System field

14.4 Buyer Fields

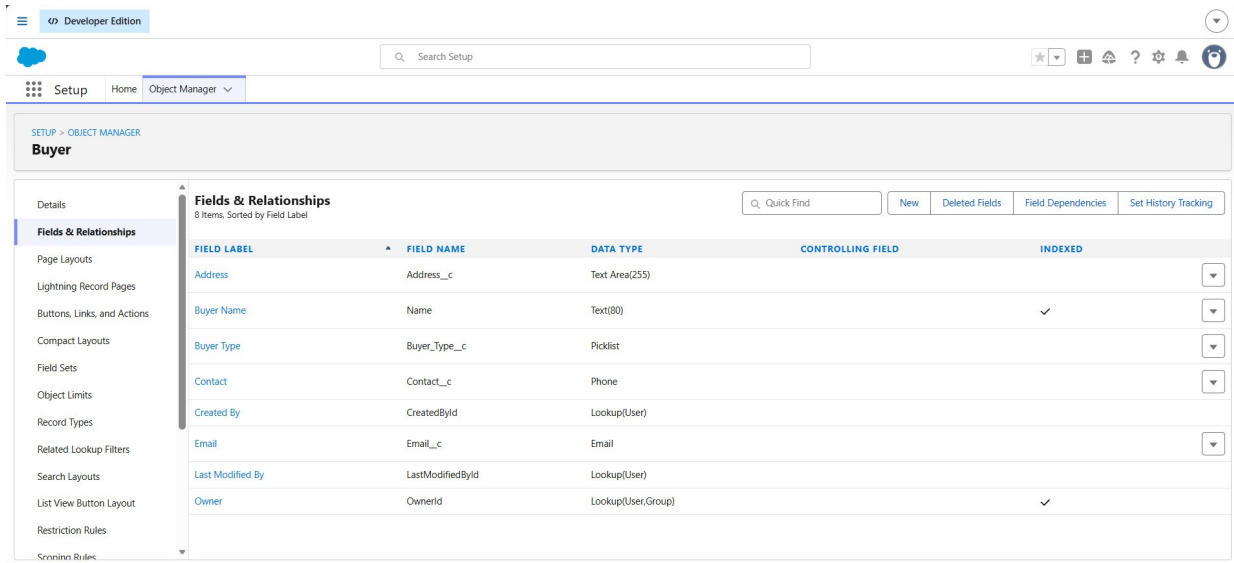


Figure 13: Buyer object: Fields & Relationships (8 items)

Table 18: Fields of Buyer__c

Field label	Field name	Data type	Notes
Buyer Name	Name	Text(80)	Record name; indexed
Buyer Type	Buyer_Type__c	Picklist	Values seen in records: Wholesale, Retail
Contact	Contact__c	Phone	
Email	Email__c	Email	
Address	Address__c	Text Area(255)	
Owner	OwnerId	Lookup(User,Group)	Indexed
Created By	CreatedById	Lookup(User)	System field
Last Modified By	LastModifiedById	Lookup(User)	System field

14.5 Dependency and Formula

Field dependency. On the Farmer object, the District picklist lists State as its controlling field (Figure 10). The New Farmer form (Figure 6) reflects this: District is disabled until a State is selected, and “View all dependencies” links are shown beneath the State and District fields. The Village picklist does not show a controlling field.

Formula field. Expected Yield per Acre on the Crop object is a Formula field with a Currency return type (Figure 12). The values in the Crop list view and in the Crops Report are consistent with the calculation $\text{Actual Yield} \div \text{Total Acres Harvested}$.

15. Lightning App Configuration

15.1 Custom Tabs

A custom object tab was created for each object so that its records can be opened from the navigation bar. Figure 14 shows the Custom Object Tabs section of Setup. The four tabs of this project are Buyers (Form icon), Crops (Bell), Farmers (Camera) and Farms (Bottle). The same list also contains a tab labelled Properties, which belongs to the org and is not part of the Farm Management System app.

Custom Object Tabs			New	What Is This?
Action	Label	Tab Style		
Edit Del	Buyers	 Form		
Edit Del	Crops	 Bell		
Edit Del	Farmers	 Camera		
Edit Del	Farms	 Bottle		
Edit Del	Properties	 Globe		

Figure 14: Custom Object Tabs in Setup listing the Buyers, Crops, Farmers and Farms tabs

15.2 App Details and Branding

The Lightning App was created in the Lightning App Builder (Figure 15). Its App Name is Farm Management System and its Developer Name is Farm_Management_System. The primary colour is #0070D2, no custom image is uploaded, and the App Launcher Preview shows the default “FM” tile.

App Settings

- App Details & Branding**
- App Options
- Utility Items (Desktop Only)
- Navigation Items
- User Profiles

App Details & Branding

Give your Lightning app a name and description. Upload an image and choose the highlight color for its navigation bar.

App Details

*App Name ⓘ

*Developer Name ⓘ

Description ⓘ

App Branding

Image ⓘ



[Upload](#)

Primary Color Hex Value ⓘ



Org Theme Options

☐ Use the app's image and color instead of the org's custom theme

App Launcher Preview



Farm Management System

Figure 15: Lightning App Builder: App Details & Branding of the Farm Management System app

15.3 Navigation Items

The Navigation Items page (Figure 16) defines the items that appear in the app navigation bar, in this order: Farms, Farmers, Crops, Buyers, Reports and Dashboards. The same items appear in the navigation bar of the running app (Figures 5 and 7 to 9).

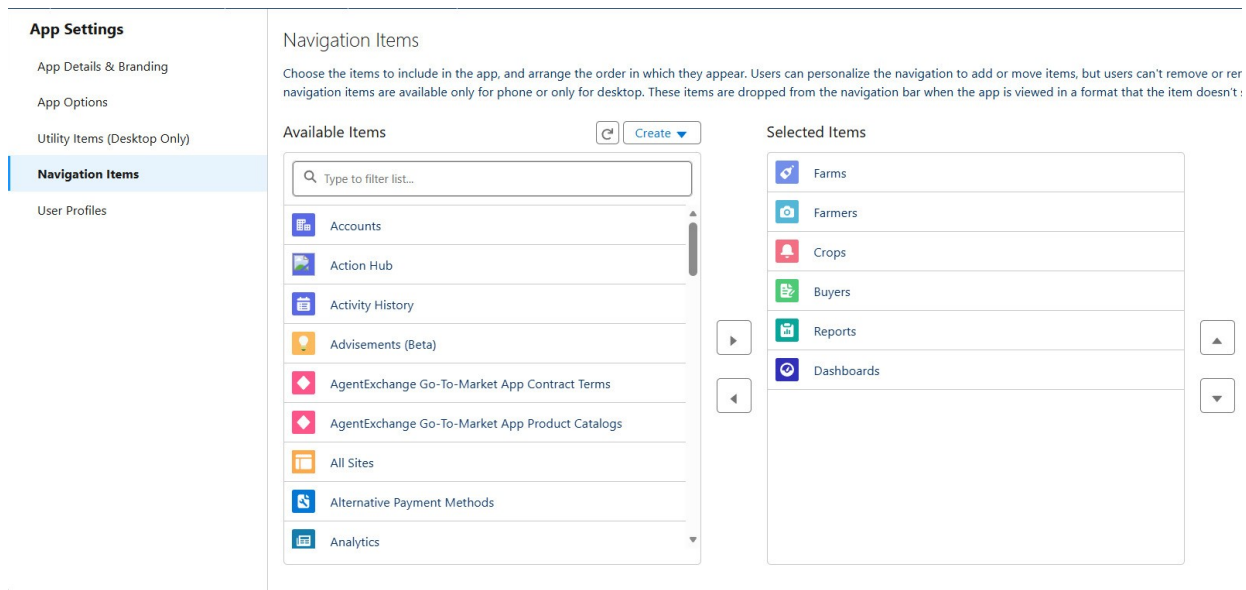


Figure 16: Lightning App Builder: Navigation Items selected for the app

15.4 User Profiles

The User Profiles page (Figure 17) determines which profiles can access the app. Three profiles are selected: System Administrator, Farm Manager and Sales Department.

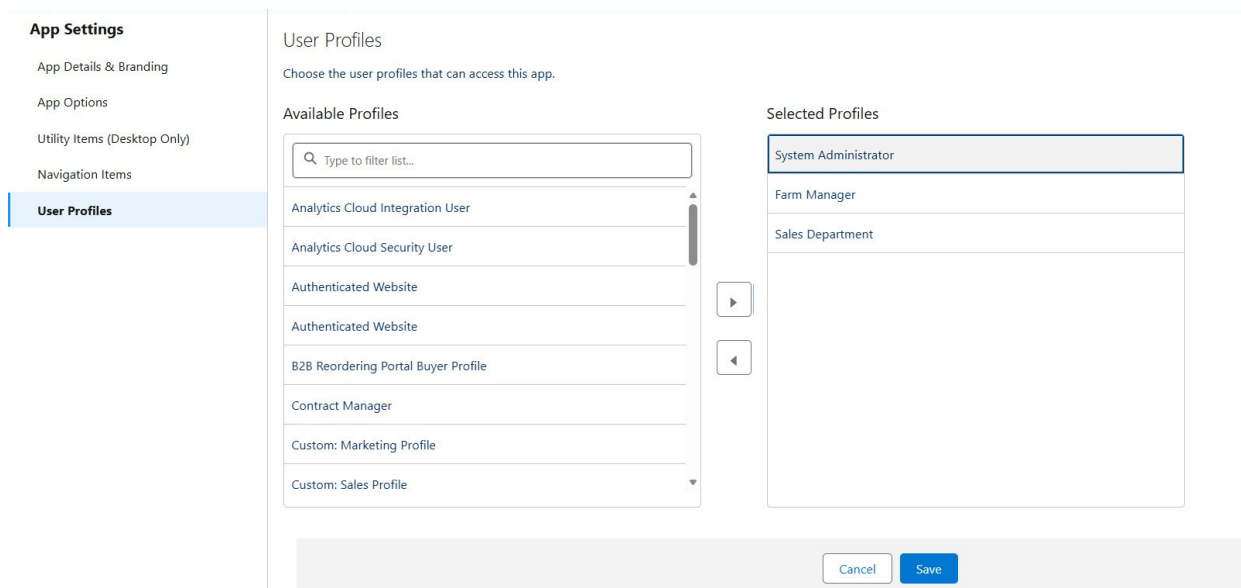


Figure 17: Lightning App Builder: profiles selected to access the app

16. Profiles, Roles, Users and Sharing Rules

16.1 Profiles

Two custom profiles, Farm Manager and Sales Department, are defined with the Salesforce user license (Figures 18 and 19). Both are marked as custom profiles and were selected for the Farm Management System app (Figure 17).

Profile Detail

Name: Farm Manager
 User License: Salesforce
 Custom Profile: ☒
 Description:
 Created By: RANJITHKUMAR P. 26/09/2026, 11:37 am
 Modified By: RANJITHKUMAR P. 26/09/2026, 5:03 pm

Buttons: Edit, Clone, Delete, View Users

Permissions: Login IP Ranges, Enabled Apex Class Access, Enabled Visualforce Page Access, Enabled External Data Source Access, Enabled Named Credential Access, Enabled External Credential Principal Access, Enabled Custom Metadata Type Access, Enabled Flow Access, Enabled Service Presence Status Access, Enabled Custom Permissions

Figure 18: Farm Manager custom profile (Profile Detail)

Profile Detail

Name: Sales Department
 User License: Salesforce
 Custom Profile: ☒
 Description:
 Created By: RANJITHKUMAR P. 26/09/2026, 11:36 am
 Modified By: RANJITHKUMAR P. 26/09/2026, 5:03 pm

Buttons: Edit, Clone, Delete, View Users

Permissions: Login IP Ranges, Enabled Apex Class Access, Enabled Visualforce Page Access, Enabled External Data Source Access, Enabled Named Credential Access, Enabled External Credential Principal Access, Enabled Custom Metadata Type Access, Enabled Flow Access, Enabled Service Presence Status Access, Enabled Custom Permissions

Figure 19: Sales Department custom profile (Profile Detail)

16.2 Roles

The role hierarchy (Figure 20) has a single top-level node with the CEO role beneath it. The Farm Manager, Sales Department, VP Development and VP Human Resources roles are placed directly below CEO. The Farm Manager and Sales Department roles are the ones used by this project.

Creating the Role Hierarchy

You can build on the existing role hierarchy shown on this page. To insert a new role, click **Add Role**.

Your Organization's Role Hierarchy

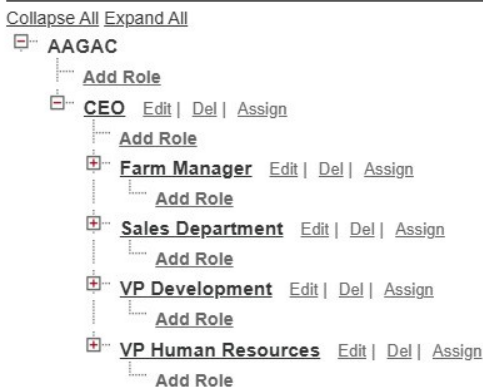


Figure 20: Role hierarchy showing the Farm Manager and Sales Department roles below CEO

16.3 Users

Two users were created, one for each function (Figures 21 and 22). Both use the Salesforce Platform user license.

Table 19: Users

User	Alias	Role	Profile	User license
Farm Manager	fmana	Farm Manager	Farm Manager	Salesforce Platform
Sales Department	sdepa	Sales Department	Sales Department	Salesforce Platform

User Detail [Edit](#) [Sharing](#) [Reset Password](#) [Freeze](#) [View Summary](#)

Name	Farm Manager	Role	Farm Manager
Alias	fmana	User License	Salesforce Platform
Email	praveenranjith914@gmail.com [Verified]	Profile	Farm Manager
Username	farm.manager.project2026@gmail.com	Active	☐
Nickname	FarmManager i	Marketing User	☐
Title		Offline User	☐

Figure 21: User detail of the Farm Manager user

User Detail [Edit](#) [Sharing](#) [Reset Password](#) [Freeze](#) [View Summary](#)

Name	Sales Department	Role	Sales Department
Alias	sdepa	User License	Salesforce Platform
Email	praveenranjith914@gmail.com [Verify] i	Profile	Sales Department
Username	sales.department.project2026@gmail.com	Active	☐
Nickname	Sales Department i	Marketing User	☐

Figure 22: User detail of the Sales Department user

16.4 Sharing Rules

Two criteria-based sharing rules appear in Sharing Settings (Figure 23). Both grant Read/Write access. No sharing rules are specified for the Farmer object in the same view.

Table 20: Sharing rules

Object	Criteria	Shared with	Access level
Crop	Crop: Crop Name EQUALS True	Role: Farm Manager	Read/Write
Buyer	Buyer: Buyer Name EQUALS True	Role: Sales Department	Read/Write

SETUP **Sharing Settings**

Work Type Group Sharing Rules [New](#) [Recalculate](#) [Work Type Group Sharing Rules Help](#) [?](#)

No sharing rules specified.

Buyer Sharing Rules [New](#) [Recalculate](#) [Buyer Sharing Rules Help](#) [?](#)

Action	Criteria	Shared With	Access Level
Edit Del	Buyer: Buyer Name EQUALS TRUE	Role: Sales Department	Read/Write

Candidate Sharing Rules [New](#) [Recalculate](#) [Candidate Sharing Rules Help](#) [?](#)

No sharing rules specified.

Crop Sharing Rules [New](#) [Recalculate](#) [Crop Sharing Rules Help](#) [?](#)

Action	Criteria	Shared With	Access Level
Edit Del	Crop: Crop Name EQUALS TRUE	Role: Farm Manager	Read/Write

Figure 23: Sharing Settings showing the Buyer and Crop sharing rules

17. Reports

Three reports are available from the Reports tab of the Farm Management System app. Each is a summary report that groups records and shows subtotals and a grand total.

17.1 Farmer with Farms Report

This report uses the “Farmers with Farms” report type. Records are grouped by Irrigation Type, and the columns are Farmer: Farmer Name, Farm: Farms Name, Contact Number, Size and Farm Type. The report returns six records: two farms with sprinkler irrigation, three with drip irrigation and one with localized irrigation (Figure 24).

Report: Farmers with Farms Farmer with Farms					
Total Records 6					
☐ Irrigation Type ↑	Farmer: Farmer Name	Farm: Farms Name	Contact Number	Size	Farm Type
☐ Sprinkler irrigation (2)	Suresh	Sunrise Farm	9843215678	8 acres	Livestock
	Selvam	Chinna Salem Farm	9894567123	7 acres	Mixed
Subtotal					
☐ Drip irrigation (3)	Kannan	Madura Agro Farm	9789012345	4 acres	Crop
	Bala	Gingee Green Farm	9887654321	5 acres	Crop
	Ramesh Kumar	Green Valley Farm	9876543210	5 acres	Crop
Subtotal					
☐ localized irrigation (1)	Murugan	Kaveri Farm	9798765432	6 acres	Mixed
Subtotal					
Total (6)					

Figure 24: Farmer with Farms report grouped by Irrigation Type (6 records)

17.2 Crops Report

The Crops Report groups records by Crop Name and shows the farm, Total Acres Harvested, Actual Yield, Expected Yield per Acre, Planting Date and Harvest date. The summary shows 5 records, 28 total acres harvested, a total actual yield of ₹1,44,000 and an expected-yield-per-acre total of ₹25,800.00 (Figure 25). Rice has two records, which together account for 10 acres.

Report: Crops Crops Report						
Total Records 5	Total Total Acres Harvested 28	Total Actual Yield ₹1,44,000	Total Expected Yield per Acre ₹25,800.00			
☐ Crop: Crop Name ↑	Farm	Total Acres Harvested	Actual Yield	Expected Yield per Acre	Planting Date	Harvest date
☐ Groundnut (1)	Madura Agro Farm	4	₹18,000	₹4,500.00	20/06/2026	25/09/2026
Subtotal		4	₹18,000	₹4,500.00		
☐ Maize (1)	Sunrise Farm	8	₹32,000	₹4,000.00	10/07/2026	20/11/2026
Subtotal		8	₹32,000	₹4,000.00		
☐ Rice (2)	Gingee Green Farm	5	₹24,000	₹4,800.00	10/06/2026	10/10/2026
	Green Valley Farm	5	₹25,000	₹5,000.00	15/06/2026	15/10/2026
Subtotal		10	₹49,000	₹9,800.00		
☐ Sugarcane (1)	Kaveri Farm	6	₹45,000	₹7,500.00	05/01/2026	05/12/2026
Subtotal		6	₹45,000	₹7,500.00		
Total (5)		28	₹1,44,000	₹25,800.00		

Figure 25: Crops Report grouped by Crop Name with totals

17.3 Buyers Report

The Buyers report groups buyers by Buyer Type and lists Buyer Name, Contact, Address and Email. It contains ten records: six Wholesale and four Retail (Figure 26).

Report: Buyers Buyers				
10				
☐ Buyer Type ↑	Buyer: Buyer Name	Contact	Address	Email
☐ Wholesale (6)	Arun Traders	📞 9876543210	Salem Tamil Nadu	aruntraders@gmail.com
	Green Harvest Buyers	📞 9798765432	Coimbatore Tamil Nadu	greenharvest@gmail.com
	Kaveri Agro Mart	📞 9789012345	Thanjavur Tamil Nadu	kaveriagromart@gmail.com
	Tamil Farmers Market	📞 9765432189	Trichy Tamil Nadu	tamilfarmersmarket@gmail.com
	Muthu Agro Traders	📞 9753124680	Villupuram Tamil Nadu	muthuagro@gmail.com
	Sunrise Food Buyers	📞 9812345670	Tiruppur Tamil Nadu	sunrisebuyers@gmail.com
Subtotal				
☐ Retail (4)	Sri Lakshmi Stores	📞 9843215678	Chennai Tamil Nadu	srilakshmistores@gmail.com
	Fresh Basket	📞 9894567123	Madurai Tamil Nadu	freshbasket@gmail.com
	Vijay Stores	📞 9887654321	Kallakurichi Tamil Nadu	vijaystores@gmail.com
	Royal Fresh Mart	📞 9867543210	Vellore Tamil Nadu	royalfreshmart@gmail.com
Subtotal				
Total (10)				

Figure 26: Buyers report grouped by Buyer Type (10 records)

18. Dashboard

The dashboard named Farmers With farms is opened from the Dashboards tab of the app (Figure 27). It contains three components, each linked to its source report through a “View Report” link:

Table 21: Dashboard components

Component	Chart type	Source report	What it displays
Farmer with Farms	Gauge (scale 0–100)	Farmer with Farms	A gauge based on the Farmer with Farms report.
Crops Report	Vertical bar chart	Crops Report	Sum of Total Acres Harvested by crop: Groundnut 4, Maize 8, Rice 10, Sugarcane 6.
Buyers	Donut chart	Buyers	Record count by Buyer Type: Wholesale 6, Retail 4, total 10.

The bar-chart and donut-chart values agree with the Crops Report and Buyers report (Figures 25 and 26). The dashboard header shows “As of 27-Sept-2026, 12:37 pm” and notes that the dashboard was last refreshed three days before the screenshot was taken, with Refresh, Edit and Subscribe buttons available.

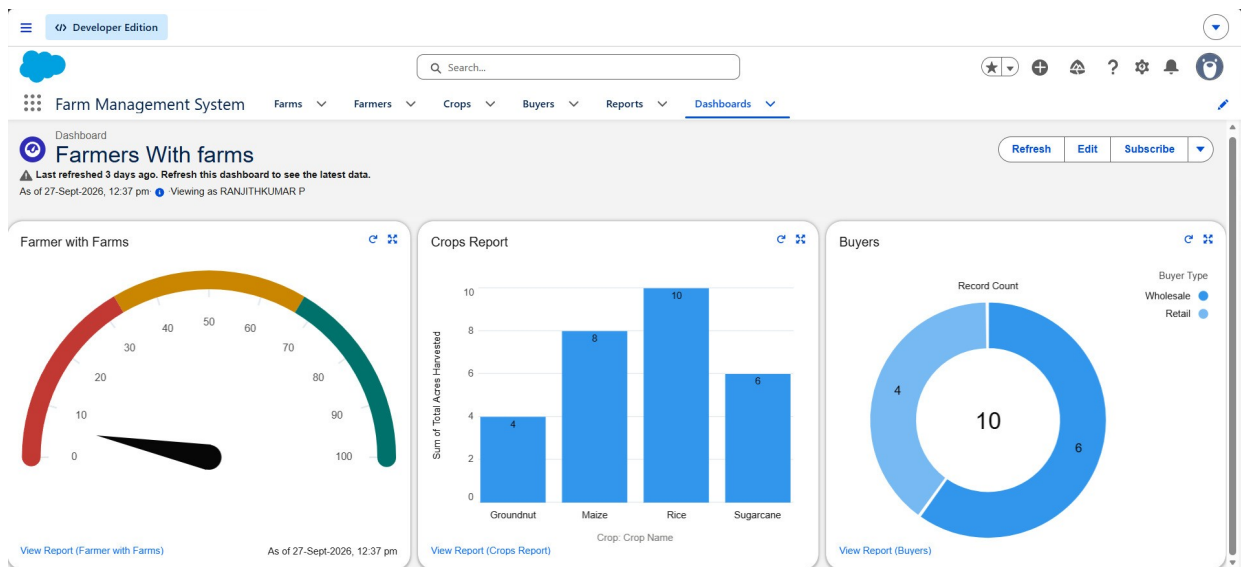


Figure 27: Dashboard “Farmers With farms” with gauge, bar chart and donut chart components

19. Testing

The system was verified through the Salesforce user interface. The checks below are limited to what the supplied screenshots demonstrate; the totals in rows T6 and T7 were additionally re-calculated from the record values shown in the list views.

Table 22: Verification checks

ID	Check	Evidence	Observed result
T1	Farmer record entry form	Figure 6	Farmer Name is marked required; the District picklist is disabled until a State is selected.
T2	Farmer records stored and listed	Figure 5	Nine Farmer records are displayed.

ID	Check	Evidence	Observed result
T3	Farm records linked to farmers	Figure 7	Six Farm records are displayed, each with a Farmer value.
T4	Crop records linked to farms; formula field	Figure 8	Five Crop records are displayed, each with a Farm value; Expected Yield per Acre equals Actual Yield ÷ Total Acres Harvested in all five.
T5	Buyer records stored and listed	Figure 9	Ten Buyer records are displayed (six Wholesale, four Retail).
T6	Report totals	Figures 24–26	Farm report: 6 records (2 + 3 + 1). Crops report: acres 4 + 8 + 5 + 5 + 6 = 28; actual yield 18,000 + 32,000 + 24,000 + 25,000 + 45,000 = ₹1,44,000. Buyers report: 6 + 4 = 10.
T7	Dashboard agrees with reports	Figures 25–27	Bar chart values (4, 8, 10, 6) and donut values (6, 4, total 10) match the reports.
T8	App access by profile	Figure 17	System Administrator, Farm Manager and Sales Department are selected for the app.
T9	Role and profile assignment of users	Figures 21–22	Each user is assigned the matching role and profile.
T10	Sharing rules configured	Figure 23	Crop rule shares with Farm Manager and Buyer rule shares with Sales Department, both Read/Write.

The supplied material does not include performance measurements, Apex tests or sign-in tests performed as the Farm Manager or Sales Department users, so no results are reported for these.

20. Advantages

- All farmer, farm, crop and buyer data is held in one Salesforce org instead of separate files.
- Related records are connected, so a crop can be traced to its farm and a farm to its farmer.
- Dependent picklists and a formula field reduce manual entry and calculation.
- The Lightning App provides a single, consistent interface with search-enabled list views.
- Reports and a dashboard summarise the data without additional tools.
- Profiles, roles and sharing rules provide controlled access.
- The application is built with declarative configuration and can be extended without custom code.

21. Limitations

The following limitations are observed from the supplied configuration:

- Buyer records are not related to farms or crops, so the system does not record which buyer purchased which produce.
- No validation rules, flows or other automation are shown in the supplied material.
- Farm Size is a Text field (for example, “7 acres”), so farm size cannot be totalled in reports.
- The Actual Yield field is a Currency field, so yield is displayed in rupees rather than in a unit of weight.
- Both sharing rules use the record Name field with the value True as criteria, which is narrow; a more meaningful field would make the rules more useful.

- Only the District field depends on State; the Village field is not dependent.
- Dashboard data must be refreshed to show the latest records, as the dashboard banner indicates.
- The application runs in a Developer Edition org and requires an internet connection; offline use is not configured.

22. Future Enhancements

The following enhancements are proposed. They are not part of the current implementation.

- Add a sales or transaction object that relates Buyer records to Crop records.
- Add validation rules, for example to require that the harvest date follows the planting date.
- Change Farm Size to a Number field so that acreage can be summed in reports.
- Make the Village picklist dependent on District.
- Add automation with Flow for crop updates and buyer notifications.
- Refine the sharing rules to use meaningful criteria fields.
- Add further reports and dashboard components, such as yield by farm type or irrigation type.
- Extend the application to mobile use and integrate external data such as weather or market prices.

23. GitHub / Source Code

The project consists of declarative Salesforce configuration made in a Salesforce Developer Edition org: custom objects, fields, relationships, tabs, the Lightning App, profiles, roles, users, sharing rules, reports and a dashboard. No custom Apex code, Lightning components or triggers are shown in the supplied material.

The source code of the project, consisting of the Salesforce configuration and metadata, is available in the following GitHub repository:

https://github.com/vp-ranjith-kumar/Farm_Management_System

24. Conclusion

The Farm Management System demonstrates how Salesforce can be configured to manage agricultural records. Four custom objects store farmers, farms, crops and buyers; relationships connect farmers to farms and farms to crops; a dependent picklist and a formula field support data entry; and a Lightning App brings the modules together with reports and a dashboard. Profiles, roles, users and sharing rules provide controlled access for the Farm Manager and Sales Department functions.

The screenshots included in this document confirm that the objects, fields, app, security settings, reports and dashboard have been configured and populated with sample data. The limitations and proposed enhancements identify clear next steps, in particular linking buyers to crops and adding validation and automation, to evolve the system into a more complete farm management solution.

25. Appendix: Screenshot Index and Sample Data

25.1 Appendix Summary

Table 23: Appendix summary

Item	Description
Source code	Salesforce configuration and metadata (repository link given in Section 23)
Dataset	Farm, Crop, Farmer and Buyer records (summarised in Section 25.3 and shown in Figures 5, 7, 8 and 9)
Tools used	Salesforce Developer Org

25.2 Screenshot Index

Every screenshot used in this document is listed below with the section in which it appears. The screenshots were taken from the project's Salesforce org. The browser window frame and the operating-system taskbar were trimmed from each capture and, where a page had large empty margins, the capture was cropped to the relevant area so that it could be shown at a larger size. The Salesforce content within each capture is unchanged.

Fig.	Content	Source file	Section
1	Farmer object details	Farmer object.png	9
2	Farm object details	farm_object.png	9
3	Crop object details	Crop object.png	9
4	Buyer object details	buyer object.png	9
5	Farmers list view	farmer-farm management system app.png	10
6	New Farmer form	farmer record adding.png	10
7	Farms list view	farm-farm management system app .png	11
8	Crops list view	crop-farm management system app.png	12
9	Buyers list view	buyer-farm management system app.png	13
10	Farmer fields	farmer Fields and Relationship.png	14.1
11	Farm fields	Farm Fields and Relationship.png	14.2
12	Crop fields	crop Fields and Relationship.png	14.3
13	Buyer fields	buyer Fields & Relationships.png	14.4
14	Custom object tabs	Tabs for all objects.png	15.1
15	App details and branding	lightning app.png	15.2
16	App navigation items	lightning app object selection.png	15.3
17	App user profiles	lightning app profiles selection.png	15.4
18	Farm Manager profile	farm manager-profile.png	16.1
19	Sales Department profile	sales department profile.png	16.1
20	Role hierarchy	roles.png	16.2
21	Farm Manager user	farm manager-user.png	16.3
22	Sales Department user	sales department-user.png	16.3

Fig.	Content	Source file	Section
23	Sharing rules	sharing rule for buyer and crop.png	16.4
24	Farmer with Farms report	farmer with farms report.png	17.1
25	Crops Report	crops report.png	17.2
26	Buyers report	buyers report.png	17.3
27	Dashboard	dashboard.png	18

25.3 Sample Data Summary

Table 24: Records visible in the supplied screenshots

Object	Records	Source
Farmer	9	Figure 5
Farm	6	Figure 7
Crop	5	Figure 8
Buyer	10	Figure 9